Muhammad Zaim Bin Rusli

Senior System Validation & Silicon Debug Engineer

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Open to relocation · Requires visa sponsorship for U.S. employment

PROFESSIONAL SUMMARY

Senior System Validation and Silicon Debug Engineer with nearly 10 years of experience spanning semiconductor manufacturing, ATE test development, pre-silicon verification, post-silicon validation, System Level Test (SLT) and data center platform qualification across Intel Xeon and AMD EPYC server processors.

Proven expertise in system-level validation, production readiness qualification, silicon bring-up, failure analysis, yield improvement, RMA debug, hardware-software-firmware integration and manufacturing test optimization. Experienced in developing automated validation infrastructures using Python and Shell scripting while driving complex cross-domain root cause analysis across silicon, firmware, BIOS, operating systems and platform hardware.

Unique background spanning wafer sort, package test, ATE, SLT, pre-silicon verification and post-silicon validation enables strong correlation across manufacturing, lab validation and field deployment environments.

SYSTEM LEVEL TEST & MANUFACTURING HIGHLIGHTS

- Nearly 10 years of experience across semiconductor manufacturing test, ATE, post-silicon validation and production qualification.
- Hands-on experience supporting High Volume Manufacturing (HVM) environments.
- Developed and maintained automated test infrastructure for validation and production readiness programs.
- Expertise correlating manufacturing failures, system-level failures and silicon root causes.
- Extensive experience debugging issues across hardware, firmware, BIOS and operating system boundaries.
- Strong background in server platform bring-up including CPU, memory, PCIe devices, NICs and BMC infrastructures.
- Experience supporting release and manufacturing decisions based on validation readiness criteria.
- Proven ability to identify validation and test coverage gaps through structured failure analysis.
- Experienced in System Level Test (SLT), Manufacturing Readiness, Production Qualification, Yield Improvement, Failure Correlation, Tester Correlation, Root Cause Analysis, RMA Analysis, Product Characterization, Performance Characterization, Platform Bring-up, Power Characterization and Automation Infrastructure development.

TECHNICAL SKILLS

Architecture & Platforms: AMD EPYC, Intel Xeon, x86 Architecture, Server Platforms, Datacenter Infrastructure, PCIe Gen4/Gen5, CXL, DDR4/DDR5, BIOS/UEFI, OpenBMC

Programming & Automation: Python, PyTest, Shell/Bash, C/C++, SystemVerilog, UVM, Simics Automation

Debug & Diagnostics: Machine Check Exceptions (MCE), WHEA, BERT, Error Injection, Status Registers, Scan Dumps, dmesg, UART, BIOS Debug, Firmware Debug

Tools: PRISM, HEXA, Git, JIRA, Simics, Verdi, DediProg, Linux, Windows

PROFESSIONAL EXPERIENCE

Functional Validation Engineer, *Advanced Micro Devices (AMD)*

Jun 2024 – Present · Penang, Malaysia

- Execute post-silicon functional validation and production readiness qualification activities on AMD EPYC server processors.
- Perform system-level validation across CPU, memory, PCIe and firmware subsystems under customer-representative workloads.
- Conduct failure analysis using Machine Check Exceptions (MCE), WHEA Logger, BERT records and hardware status registers to identify silicon and platform root causes.
- Execute error injection, reliability testing and platform characterization to evaluate robustness of production silicon.
- Support platform bring-up, hardware assembly, BIOS configuration, OpenBMC management, operating system provisioning and regression infrastructure maintenance.
- Develop Python-based automation solutions within the PRISM validation framework to improve validation throughput and execution consistency.
- Work cross-functionally with architecture, firmware, BIOS and platform engineering teams to resolve complex system-level issues and improve validation coverage.

System Validation Engineer, *Intel Microelectronics* Aug 2019 – May 2024 · Kulim, Malaysia

Post-Silicon System Validation

- Executed structured test plans, silicon qualification activities and manufacturing readiness validation on Intel Xeon server processors.
- Defined system-level validation methodologies supporting production release decisions for complex server SoCs.
- Developed and maintained Python and Shell-based automation within the HEXA validation framework to improve regression efficiency and validation scalability.
- Conducted extensive stress, corner-case and concurrency testing to characterize silicon quality, reliability and scalability under production-like conditions.
- Performed root cause analysis across hardware, BIOS, firmware and operating system layers to resolve manufacturing-blocking and customer-impacting issues.
- Validated security, trusted execution and platform initialization features including TDX, SGX, ACM and SEAM Loader integration.

- Executed large-scale concurrency testing involving hundreds of simultaneous workloads to validate platform stability and memory resource management.
- Collaborated with architecture teams to drive silicon defect identification, debug and closure.
- Supported manufacturing release criteria through objective silicon validation metrics and readiness assessments.

Pre-Silicon Validation & Emulation

- Executed SoC functional validation in Simics Virtual Platform and FPGA emulation environments.
- Developed automation to validate boot flows, feature enablement and register-level functionality.
- Validated security and trusted execution features prior to silicon availability, reducing downstream silicon debug effort.

Client & Embedded Validation

- Performed system-level validation on embedded and client platform SoCs.
- Developed automation supporting boot verification and platform readiness checks.
- Collaborated with firmware and software teams to resolve integration issues and improve platform stability.

IP Verification Engineer, *Intel Microelectronics* Mar 2018 – Aug 2019 · Penang, Malaysia

- Verified R-Link Soft IP functionality using SystemVerilog and UVM methodologies.
- Developed constrained-random test scenarios to improve feature coverage and RTL bug detection.
- Performed waveform analysis and root cause investigation using Verdi simulation tools.
- Developed assertions and coverage metrics supporting verification closure objectives.
- Collaborated closely with RTL design teams to resolve protocol and integration issues prior to tape-out.

Test & Product Engineer, *Malaysia Microelectronic Solutions (MMS)*
Aug 2017 – Mar 2018 · Selangor, Malaysia

- Developed and debugged production ATE test programs supporting semiconductor high-volume manufacturing.
- Performed yield analysis, electrical characterization and probe-level failure debug.
- Supported manufacturing test optimization and production release readiness.
- Collaborated with external organizations to resolve hardware and probe-related production issues.

Graduate Engineer, *NXP Semiconductors* Sep 2016 – Jul 2017 · Selangor, Malaysia

- Developed Teradyne J750 production test solutions.
- Performed device characterization, failure analysis and tester correlation activities.
- Supported production qualification and yield improvement initiatives.
- Maintained automation scripts supporting manufacturing and characterization activities.

EDUCATION

B.Eng. (Hons.), Electronic Engineering, *Universiti Teknologi MARA (UiTM)* 2016